Points of significance

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Errors in predictor variables

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It is the mark of an educated mind to rest satisfied with the degree of precision that the nature of the subject admits and not to seek exactness where only an approximation is possible.
—Aristotle

We've already seen how regression can estimate the influence of one or more variables (predictors) on the distribution of another variable (response)[1–3]. In linear regression[1], we estimate the mean of the response at each combination of predictors. In quantile regression[2], we estimate quantiles, which can tell us about the probability distribution of the response. Finally, in logistic regression[3], we estimate the class probabilities of a binary response variable. In all these, the predictors are assumed to have known values that are measured without error. Practically, however, predictors are often measured with error. This has a profound (but predictable) effect on the estimates of relationships among variables—the so-called 'error in variables' problem.

Error in measuring the predictors is often ignored. This can be justified when the goal is to predict Y from the observed predictors W, or when the measurement error is small compared to the range of the true predictor X. However, when the objective is to understand the relationship between Y and X and the error in X is large, large biases and additional variance may be introduced, which, in turn, can affect estimates of other covariate effects.

Suppose we seek to determine the effect of predictor X on the response Y with sample size n. In ordinary linear regression[1] we would write $Y=\beta_0+\beta_1X+\varepsilon$ and estimate the unknown intercept β_0 by $b_0 = \bar{y} - b_1\bar{x}$ and the unknown slope β_1 by $b_1 = s_{XY}/s_{XX}$. Here $\bar{x}$ and $\bar{y}$ are the sample means of X and Y, respectively, $s_{XY} = \mathrm{Cov}(X,Y) = \sum(x_i-\bar{x})(y_i-\bar{y})/(n-1)$ is the sample covariance between X and Y, and $s_{XX} = \mathrm{Var}(X) = \sum(x_i-\bar{x})^2/(n-1)$ is the sample variance of X. Finally, ε is the error in Y, assumed independent of X, with mean 0 and variance σ_ε^2.

The relationship between the true (β_1) and estimated (b_1) slope can be derived from

$$\begin{aligned} s_{XY} &= \sum(x_i-\bar{x})\left(\beta_0+\beta_1x_i+\varepsilon_i-\right.\\ &\quad\left.[\beta_0+\beta_1\bar{x}+\bar{\varepsilon}]\right)/(n-1)\\ &= \beta_1 s_{XX} + s_{X\varepsilon}, \end{aligned}$$

which yields $b_1 = \beta_1 + s_{X\varepsilon}/s_{XX}$. Because X and ε are independent, the second term has zero expectation, written as $\mathrm{E}(s_{X\varepsilon}/s_{XX}) = 0$. Thus, while b_1 varies from sample to sample, it is an unbiased estimator—it will vary around the true value but will not (on average) be higher or lower than β_1. It is important to note that ε is the variability of Y around its mean on the regression line for the corresponding value of X—it is not the error in X.

Actual error in X can cause (potentially substantial) bias in b_1 as well as increased variability of the estimate. This is the crux of the errors in variables problem: because we (generally) do not directly observe X, but a noisy version of it, W, our slope estimate, will (on average) be systematically off. The underlying cause of this may be measurement error in X or the fact that X is actually a long-term average and our measurement, W, is an instantaneous value (for example, blood glucose, caloric intake or blood pressure).

To model this error, suppose that instead of X we observe $W=X+\omega$, where the noise ω is independent of both X and ε, with zero mean and variance τ^2. This is the classical measurement error model (the simplest of many), applicable when the noisy measurements are centered on their corresponding true values with errors that are independent of the true value.

Let's look at the case of a single noisy predictor X, with mean μ_X and variance σ_X^2. Since X is not measured, we cannot regress Y on X (written as $Y\sim X$) but only on W ($Y\sim W$). In this regression, the estimated slope a_1 is then computed with W replacing X in the definitions above, yielding

$$\begin{aligned} a_1 &= s_{WY}/s_{WW}\\ &= (s_{XY}+s_{\omega Y})/(s_{XX}+s_{\omega\omega})\\ &= \beta_1 s_{XX}/(s_{XX}+s_{\omega\omega}+2s_{\omega X})\\ &\quad + (s_{\omega Y}+s_{X\varepsilon})/(s_{XX}+s_{\omega\omega}+2s_{\omega X}). \end{aligned}$$

Because X, ω and ε are independent, $\mathrm{E}(s_{\omega X}) = \mathrm{E}(s_{\omega Y}) = \mathrm{E}(s_{X\varepsilon}) = 0$ and we can write the expected value of a_1 as

$$\begin{aligned} \mathrm{E}(a_1) &= \beta_1\sigma_X^2/\sigma_W^2\\ &= \beta_1\sigma_X^2/(\sigma_X^2+\tau^2)\\ &= \beta_1/(1+\tau^2/\sigma_X^2), \end{aligned}$$

where σ_X^2 is the variance of X.

Thus, when X is noisy ($\tau^2>0$), we have slope attenuation bias: a_1 has the same sign but (on

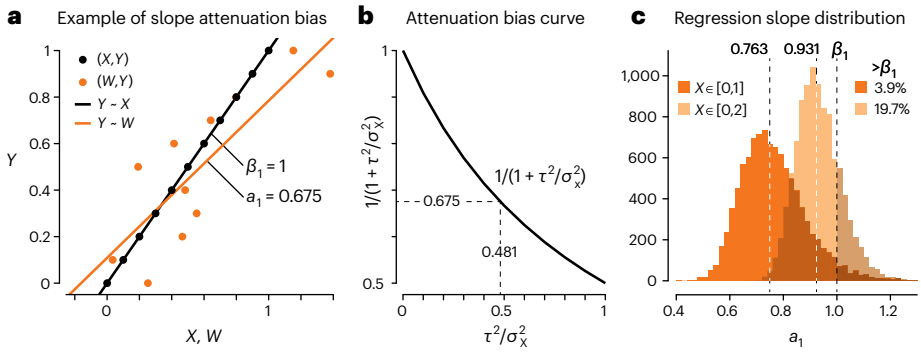

Fig. 1 | Noise in the predictor variable generally reduces the magnitude of the slope estimate in simple linear regression. a, Regression of the ground truth $Y=X$ (black line, slope $\beta_1=1$) and the regression $Y\sim W$ where $W=X+\omega$ is the noisy version of X with ω distributed normally with zero mean and variance of $\tau^2=0.04$ (orange line, slope $a_1=0.675<\beta_1$). **b**, The approximate profile of the attenuation bias as a function of the relative noise (τ^2/σ_X^2) in the predictor. Dashed lines show relative noise and bias factor based on the sample variances $s_{\omega\omega}=0.053$ and $s_{XX}=0.11$ in **a**. **c**, The distribution (dark orange) of slope estimates a_1 of 10,000 iterations of the regression in **a**. The mean of the distribution is 0.763, with 3.9% of slopes being overestimates. Also shown is the slope distribution (light orange) when the range of X is doubled.

average) smaller magnitude than the true slope ($|E(a_1)| < |\beta_1|$). In other words, the relationship between the noisy predictor and the response will (on average) underestimate the relationship between the true (noiseless) predictor and the response. As τ^2 (the noise in X) increases relative to σ_X^2 (the variance of X), the bias may become unacceptable. For example, if $\tau^2/\sigma_X^2 = 0.5$, the attenuation is $1/(1+0.5) = 0.67$; then $E(a_1)$ is only 67% of β_1. Conversely, the bias decreases as σ_X^2 increases relative to τ^2.

To estimate the bias in a given linear regression, we need to know σ_X^2 and τ^2, which could be estimated using s_{XX} and $s_{\omega\omega}$, respectively. However, because we do not measure X directly, we only have access to $s_{WW} = s_{XX} + s_{\omega\omega} + 2s_{X\omega}$ and (generally) cannot untangle s_{XX} from $s_{\omega\omega}$. When X and ω are uncorrelated, their sample covariance $s_{X\omega}$ is (typically) small and can be ignored.

Nevertheless, we may be able to estimate τ^2 if, for example, the noise in X is due to known measuring error. Alternatively, we may be able to take replicate measurements of W (for example, W_1, W_2) for some subjects, allowing us to estimate both σ_X^2 and τ^2, since $E(\mathrm{Cov}(W_1, W_2)) = \sigma_X^2$ and $E((W_{1i} - W_{2i})^2) = E(\omega_{1i} - \omega_{2i})^2 = 2\tau^2$. Finally, in some scenarios (for example, food diaries, lab measurements), it may be possible to obtain the true value of X_i for a random portion of the sample by using a more accurate (but presumably more expensive) measuring device that allows us to estimate σ_X^2 from X_i and τ^2 from $\omega_i = W_i - X_i$.

To understand this concretely, suppose $Y = X$ is the ground truth relationship (black line, Fig. 1a). Now, instead of X (black dots), suppose that we measure its noisy version (W, orange dots), here generated from X by adding Gaussian noise with zero mean and $\tau^2 = 0.04$. A linear regression of $Y \sim W$ (orange line) yields a biased slope of $a_1 = 0.675$. Because we've simulated both X and W, we know the actual sample variances ($s_{\omega\omega} = 0.053$, $s_{XX} = 0.11$) and can estimate the bias factor directly using $1/(1 + 0.053/0.11) = 0.675$ (Fig. 1b).

By performing this simulation 10,000 times, we can empirically determine the distribution of a_1 (Fig. 1c). The estimated slope will be (on average) substantially smaller (0.763) than the ground truth ($\beta_1 = 1$), though in a small number of samples (3.9%) it will be larger. In this example, all of the variability in a_1 (that is, the spread of the distribution) is due to the noise in X, since there is no noise in Y ($b_1 = \beta_1$). If Y were noisy, b_1 would also come from a distribution, but with $\mathrm{Var}(a_1) > \mathrm{Var}(b_1)$.

One quick way to reduce attenuation bias is to increase s_{XX} relative to τ^2, which can be achieved by increasing the range of X. For example, if we double the range of X and use $X = (0, 0.2, 0.4, \ldots 2)$ instead of $(0, 0.1, 0.2, \ldots 1)$, we increase s_{XX} from 0.11 to 0.44 and mitigate the attenuation bias (on average) from 0.763 to 0.931 (dark orange histogram, Fig. 1c). Unfortunately, this is not always possible, and attempts to include more extreme values of X may lead to the introduction of high leverage points[4] in regions where the linear model may not be valid.

What are the practical consequences of this attenuation? If our study collects (W, Y) data pairs in order to predict Y from W, then the attenuation is irrelevant. The linear predictor $\hat{Y} = a_0 + a_1 W$ is the best linear predictor for some future observation of W. However, the intersample variability in the regression coefficients (a_0, a_1) is higher when using W as a predictor than in the corresponding coefficients (b_0, b_1) if X had been used. And, given that the magnitude of a_1 is attenuated and $\mathrm{Var}(a_1) > \mathrm{Var}(b_1)$, a test that the slope is zero has reduced power. As well, any confidence and prediction intervals around the regression line are larger.

The bias in the slope becomes very relevant when we try to use our estimate a_1 (the rate at which Y varies with W) to understand β_1 (the rate at which Y varies with X). For example, average blood glucose, X, is a complex quantity that cannot be measured quickly and directly[5]. Most often it is approximated by blood glucose, W, a proxy subject to large fluctuations (even under controlled conditions).

Making predictions from a regression on W is not straightforward. For example, if we find that a 10% increase in blood glucose (W) results in a 20% increase in a response variable (Y), we cannot assume the same increase in Y for a 10% increase in average blood glucose (X).

From our demonstration of bias, we know that it is very likely that the increase in the response Y will be larger than 20% – possibly much larger if τ^2 is large. To estimate the increase in Y, we would need to perform an attenuation correction. But, without knowing τ^2/σ_X^2 accurately, our correction would merely be an educated guess. A relatively easy way to estimate this ratio is to collect replicates of blood glucose from the same patients on days far enough apart that fluctuations are independent.

To illustrate how replicates can be used, let's continue with our example from Fig. 1. Let's consider the values of W in Fig. 1a to be the first set of replicates (W_1, orange dots, Fig. 2a) and

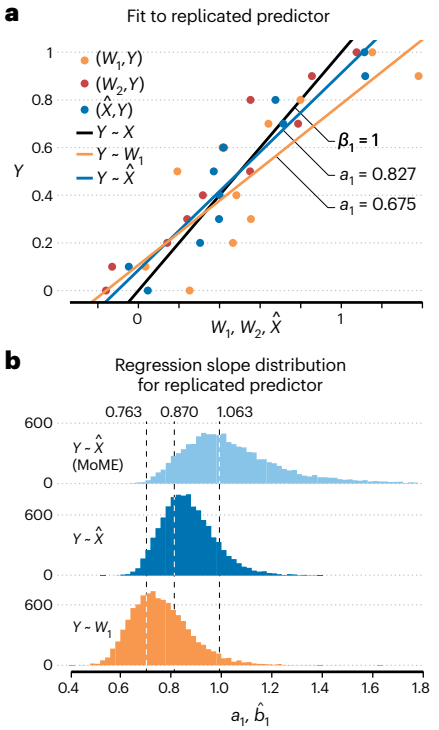

Fig. 2 | Slope attenuation bias correction using replicates of *W*. a, Averaging two (or more) replicates of W (W_1, W_2) provides an estimate ($\hat{X}$, blue dots) of X and a regression $Y \sim \hat{X}$ (blue line) that is substantially closer ($a_1 = 0.827$, blue line) to the ground truth (black line) than $Y \sim W_1$ ($a_1 = 0.675$, orange line). **b**, Distribution of slope estimates from $Y \sim W_1$ (orange histogram) as well as $Y \sim \hat{X}$ (dark blue histogram), further corrected with method of moments (MoME) (light blue histogram).

similarly simulate a second set (W_2, red dots, Fig. 2a) using the same variance $\tau^2 = 0.04$. The average of each replicate pair (blue dots, Fig. 2a) is an unbiased estimate $\hat{X}_i$ of X_i. If we regress $Y \sim \hat{X}$ (blue line, Fig. 2a) we obtain a slope $a_1 = 0.827$, which is already closer to the ground truth (black line, Fig. 2a) than the slope estimate with only one measurement of X_i ($a_1 = 0.675$, orange line, Fig. 2a). Indeed, a distribution of slopes from replicate pairs (dark blue histogram, Fig. 2b) illustrates that we can (on average) reduce the attenuation bias by a factor of about 2 (from 23.7% to 13%).

We can further correct the slope based on replicates by observing that, because ω_1 and ω_2 are independent, $\mathrm{Cov}(W_1, W_2)$ can be considered an estimate of σ_X^2 and $\mathrm{Var}(\hat{X}_i)$ an estimate of $\sigma_{\hat{X}}^2$. With this, we can use the method of moments estimator (MoME) and refine our estimate of b_1 by $\hat{b}_1 = a_1 \mathrm{Var}(\hat{X}_i)/\mathrm{Cov}(W_1, W_2)$. This has the effect of reducing the attenuation bias by another factor of two (from 13% to 6.3%)

but substantially increases the variability in our corrected slope (light blue histogram, Fig. 2b).

If we were to have replicates for only a random subsample, we would first compute a_1 by regressing $Y \sim W_1$ and then apply MoME correction using $\hat{b}_1 = a_1 \mathrm{Var}(W_1)/\mathrm{Cov}(W_1, W_2)$. Here, the variance is computed only from W_1 and the covariance is computed only from observations with both W_1 and W_2.

In more complex regression settings (multiple linear regression, quantile regression, logistic regression), the theoretical computations are more difficult and depend on the correlation between the noisy predictor(s) and any additional noise-free predictors. Even in simple linear regression, the effects of error in the predictor are not necessarily straightforward. For example, if the noisy predictor is biased or the errors in the predictor are correlated with the response, attempts at correction may increase the slope bias, rather than attenuating it[6]. We will consider some of these scenarios as well as more sophisticated bias corrections in a follow-up article.

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Competing interests

The authors declare no competing interests.